

Audit Trail — Case 11: Shoulder closure on undivided highway

Field	Value
MUTCD typical application	TA-3
CDOT standard sheet	S-630-1
Case	Case 11: Shoulder closure on undivided highway
Taper length	54 ft (L/3 (shoulder taper))
Buffer space	250 ft
Device spacing — taper	35 ft
Device spacing — tangent	70 ft
Crew steps	11
Routing	shoulder_no_reduction

Taper Length

Speed 35 mph < 45 mph threshold -> using $L = W \times S^2 / 60$

$$L = 8 \times 35^2 / 60 = 163 \text{ ft}$$

$$L/3 = 163 / 3 = 54 \text{ ft}$$

Field	Value
Closure type	shoulder
Offset (shoulder width)	8 ft
Required taper (L/3 (shoulder taper))	54 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-3. Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure, applied to undivided highway)

Buffer Space

MUTCD Table 6B-2: 250 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

35 mph -> 35 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

35 mph -> 70 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

$L/3 = 54 \text{ ft} / 35 \text{ ft max spacing} \rightarrow 4 \text{ drums at } 18.1 \text{ ft intervals}$

$800 \text{ ft} / 70 \text{ ft max spacing} \rightarrow 13 \text{ cones at } 66.7 \text{ ft intervals}$

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 13 tangent + 2 downstream = 15 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 35 mph, road_type = 'urban_low' -> Table 6B-1 category: urban_low

A = 100 ft, B = 100 ft, C = 100 ft

Position	Code	Station (ft)	Distance from Taper (ft)
C (furthest)	W20-1	1,404	300 upstream
B (middle)	W20-2	1,304	200 upstream
A (nearest)	W21-5aR	1,204	100 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: False. Signs placed: 0 left, 8 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 1,404 ft. Required: 1. Placed: 1.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 35 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: True

CDOT Case Reference

Case 11: Shoulder closure on undivided highway

This scenario follows CDOT Standard Plan S-630-1, Case 11 — the general shoulder-work typical (drawn for freeway/expressway) — applied to an undivided highway with single-side signing per CDOT S-630-1 Sheet 2 General Note 8.

The generated plan follows the same device layout as the reference case with spacing computed for 35 mph.

Routing: shoulder_no_reduction

Reference:

[https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20\(19-Page%20Set\).pdf](https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf)

Geometry Validation

Work zone 800 ft vs required taper 54 ft (L/3 (shoulder taper)) + buffer 250 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

Corridor check not run (no site coordinates supplied).

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6C.02 — pedestrian considerations in work zones	Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

0 reference(s) pending verification. Tracking: <https://github.com/rtmakatura/conestruct/issues/19>